after t_1 sec and the second echo after t_2 sec. The distance between the two hills is

a) $\frac{v(t_1 - t_2)}{2}$ b) $\frac{v(t_1 t_2)}{2(t_1 + t_2)}$

c) $v(t_1 + t_2)$ d) $\frac{v(t_1 + t_2)}{2}$

8. An air column in a pipe which is closed at one end, will be in resonance with the vibrating body of frequency 83Hz. Then the length of the air column is

a) 1.5 m (b) 0.5 m

(c) 1.0 m (d) 2.0 m

9. The displacement y of a wave travelling in the x direction is given by $y = (2 \times 10^{-3}) \sin(300t - 2x + \frac{\pi}{4})$, where x and y are measured in metres and t in second. The speed of the wave is

(a) 150 ms^{-1} (b) 300 ms^{-1}

(c) 450 ms^{-1} (d) 600 ms^{-1}

10. Consider two uniform wires vibrating simultaneously in their fundamental notes. The tensions, densities, lengths and diameter of the two wires are in the ratio 8 : 1, 1 : 2, x : y and 4 : 1 respectively. If the note of the higher pitch has a frequency of 360 Hz and the number of beats produced per second is 10, then the value of x : y is

(a) 36 : 35

(b) 35 : 36

(c) 1 : 1

(d) 1 : 2

11. Which of the following represents a wave

(a) $(x - vt)^3$ (b) $x(x+vt)$

(c) $\frac{1}{(x + vt)}$ (d) $\sin(x+vt)$

12. A man sitting on a swing which is moving to an angle of 60° from the vertical is blowing a whistle which has a frequency of 2.0 k Hz. The whistle is 2.0 m from the fixed support point of the swing. A sound detector which detects the whistle sound is kept in front of the swing. The maximum frequency the sound detector detected is

(a) 2.027 *k*Hz (b) 1.974 *k*Hz

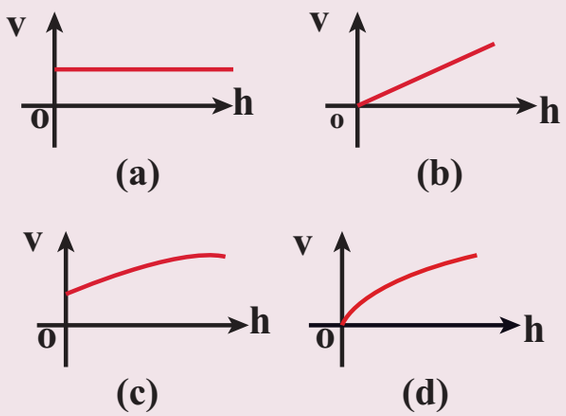
(c) 9.74 *k*Hz (d) 1.011 *k*Hz

13. Let $y = \frac{1}{1 + x^2}$ at t = 0 s be the amplitude of the wave propagating in the positive x-direction. At t = 2 s, the amplitude of the wave propagating becomes $y = \frac{1}{1 + (x-2)^2}$. Assume that the shape of the wave does not change during propagation. The velocity of the wave is

(a) 0.5m s^{-1} (b) 1.0m s^{-1}

(c) 1.5m s^{-1} (d) 2.0m s^{-1}

14. A uniform rope having mass m hangs vertically from a rigid support. A transverse wave pulse is produced at the lower end. Which of the following plots shows the correct variation of speed v with height h from the lower end?



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